Lesson 1-6 • Average Rate of Change

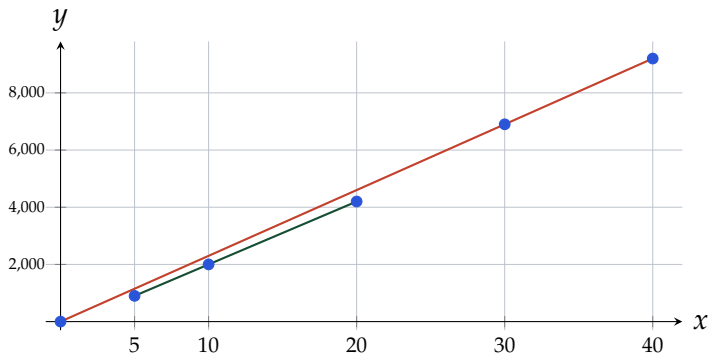
Notes (filled in)

Objective: find and interpret the average rate of change on an interval, with units.

Moussa's bike ride

Moussa rides his bike along the Corniche in Dakar. The distance he has ridden, in meters, t minutes after he starts, is $D(t)$ for $0 \le t \le 40$.

t (min)	0	5	10	20	30	40
$D(t)$ (m)	0	900	2000	4200	6900	9200



Average rate of change

The average rate of change of f on $a \le x \le b$: $\dfrac{f(b)-f(a)}{b-a} = \dfrac{\Delta y}{\Delta x}$

Graphically, it is the slope of the secant line through the two points $(a, f(a))$ and $(b, f(b))$.

Units: output units per input unit .

Example 1

a) Sketch the line that represents the average rate of change of D over the whole interval.
b) Find the average rate of change of D on $0 \le t \le 40$. Give units.

a) The red line from $(0,0)$ to $(40,9200)$. b) $\dfrac{D(40)-D(0)}{40-0} = \dfrac{9200}{40} = 230$ meters per minute.

Example 2

a) Sketch the line that represents the average rate of change of D on $5 \le t \le 20$.
b) Find it and interpret it in context.

a) The green line from $(5,900)$ to $(20,4200)$. b) $\dfrac{4200-900}{20-5} = \dfrac{3300}{15} = 220$ m/min: between minute 5 and minute 20, Moussa rode on average 220 meters per minute.

Example 3	Find the average rate of change of $f(x) = x^2 - 2x$ on $[1, 4]$. $f(4) = 8, f(1) = -1$: $\dfrac{8-(-1)}{4-1} = \dfrac{9}{3} = 3$.
Summary	Average rate of change $= \frac{f(b)-f(a)}{b-a}$, the slope of the secant line between two points. Always give units: output units per input unit (meters per minute).